SMART INDIA HACKATHON 2026

Idea Submission — Problem Statement Details

PS ID

26016

Problem Statement ID	**26016**
Problem Statement Title	**Real-Time National Land Acquisition & Management System for End-to-End Digital Monitoring and Decision Support**
Theme	**Smart Automation**
PS Category	**Software**
Team ID	<Team ID — as registered on SIH portal>
Team Name	<Team Name — as registered on SIH portal>
Organization / Department	**Ministry of Rural Development · Dept. of Land Resources (DoLR)**

Proposed by: <Team Name> · <College / Institute Name>

Do not forget to replace <placeholders> before export!

IDEA TITLE

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

NLAMS — a GIS-powered digital twin of the complete land acquisition lifecycle (Telangana prototype, national-ready)

Detailed explanation of the proposed solution

- Web platform — digitises the complete lifecycle — Proposal → Scrutiny → Notification → Verification → Award → Possession → Compensation, as per RFCTLARR Act, 2013.
- GIS core — real boundaries of all 33 Telangana districts as vector maps; each project drawn as a route alignment with its land parcels on it.
- Stage-native parcels — every parcel carries its live legal stage (Notified / Verified / Awarded / Possession / Compensated / Disputed), colour-coded on map.
- Role-based access — Central / State / District / Field views; every action audited.
- Compensation ledger — DBT-style per-family tracking with SLA clocks & one-click release.
- Alerts & risk engine — projects scored 0–100; breaches raise alerts with recommended actions.

How it addresses the problem

- Single source of truth — projects, parcels, families & payments in one system — no silos across Revenue, R&B, I&CAD, TSIIC.
- Real-time monitoring — stage-wise progress live on dashboards & map — delays surface in hours, not months.
- Spatial intelligence — map shows exactly where acquisition is stuck (village / survey no.) — impossible in registers & PDFs.
- Faster compensation — per-family SLA alerts stop cases crossing 90+ days.
- Transparency for families — their parcel status & payment date on demand — trust, fewer protests.

Innovation and uniqueness

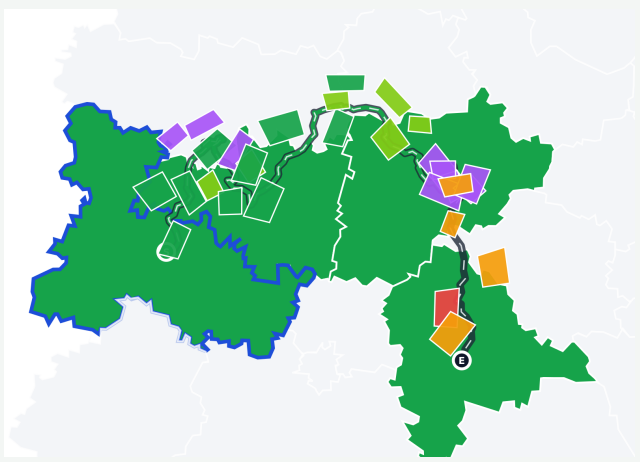
- Alignment-corridor GIS — parcels strung along the actual route (RRR arc, Warangal ORR ring) in workflow colours — not dots on a map.
- Legal-workflow-native — Sec 4/11/26/30/38 of the 2013 Act map 1:1 to screens & actions — an officer file becomes a workflow.
- Zero-infrastructure deploy — runs offline as a single file in any browser — works in low-connectivity offices; no licence cost.
- Stage-click filtering — click a pipeline stage on the map → only parcels at that stage highlight.
- State-agnostic engine — swap boundaries & records → same system for any state; scales to a national grid for DoLR.

TECHNICAL APPROACH

Technologies used

- Prototype (working today) — HTML5 · CSS3 · Vanilla JS · in-house SVG GIS engine · Python GeoJSON→SVG pipeline. Zero dependencies, offline-capable.
- Production stack (planned) — React + TypeScript · Node.js/NestJS APIs · PostgreSQL + PostGIS · Redis · object storage for documents.
- Government integrations — Dharani (land records) · Bhuvan/NIC GIS layers · CFMS (DBT payments) · e-Gazette via secure API gateway.
- Smart automation — rule-engine SLA alerts · payment anomaly checks · AI risk-scoring roadmap (delay-pattern ML).

Working prototype — screenshots



Route-alignment GIS: RRR Northern Arc — corridor parcels in workflow colours, green = land acquired

Methodology — implementation flow

Ingest boundaries & records → **Digitise lifecycle (RFCTLARR stages)** → **Map parcels to route corridors** → **Monitor · Alert · Action** → **DBT compensation & audit trail**

- Process — requirements from the Act → data model → GIS engine → workflow module → integration APIs → one-district pilot → state rollout.
- Prototype status — fully interactive end-to-end system already built — approve, verify, publish & pay all work live in the demo.

State dashboard map: live district status across all 33 districts of Telangana

FEASIBILITY AND VIABILITY

Analysis of the feasibility of the idea

✓ Already built

working end-to-end prototype — not a concept

✓ Legal alignment

digitises the existing RFCTLARR 2013 process — no policy change needed

✓ Open data ready

boundaries & geoportals are public; records APIs are govt-owned

✓ Zero-cost pilot

single-file tech runs in any district office browser

✓ DILRMP synergy

complements the national land-records modernization programme

Potential challenges & risks — and strategies to overcome them

CHALLENGE / RISK	MITIGATION STRATEGY
Integration with Dharani / CFMS legacy systems	→ Standard REST/CSV fallback formats; phased read-only integration first; DoLR-led API gateway under Digital India framework
Field data quality (GPS surveys, owner records)	→ Mobile app with offline capture, mandatory validation rules, cross-check with Dharani before stage promotion
Adoption & change management for officers	→ Role-simple UI (3 clicks per action); training modules; state SOP notification; pilot district first
Security & privacy of land & family data	→ RBAC (4 levels) · encryption in transit & at rest · complete audit trail · DPDP Act 2023 compliance by design
National scale (36 States/UTs)	→ State-agnostic engine — boundaries, categories & rates are configuration, not code

IMPACT AND BENEFITS

Potential impact on the target audience

Affected families

See their parcel's exact stage & payment date — dignity, predictability, faster money in hand

Field & district officers

3-click actions, GPS-verified parcels, SLA clocks — files move without files

State & Central Govt

Live corridor-level visibility; early alerts prevent escalations & litigation

Citizens at large

Highways, irrigation & industry start on time — growth not stuck in land disputes

Benefits of the solution

SOCIAL

- Transparency ends information asymmetry between State and families
- Fewer disputes & protests — less displacement distress
- Telugu UI + SMS/WhatsApp status (roadmap) for affected families
- Full audit trail = fair, defensible awards

ECONOMIC

- Faster commissioning of highways, irrigation & industrial corridors
- Months saved on mega-projects avoid crores in cost escalation
- DBT-linked ledger plugs leakage; accurate budgeting from live data
- Lower litigation spend for the exchequer

ENVIRONMENTAL / GOVERNANCE

- Early forest & land-overlap screening (FCA) before alignment freeze
- Digitised records cut paper, travel & re-surveys
- Planning data avoids re-alignments that waste acquired land
- Reusable national asset for DoLR — one engine, every state

RESEARCH AND REFERENCES

Details / Links of the reference and research work

1	**RFCTLARR Act, 2013 — Right to Fair Compensation & Transparency in Land Ac…**	Dept. of Land Resources, MoRD — dolr.gov.in · indiacode.nic.in
2	**Digital India Land Records Modernization Programme (DILRMP)**	dolr.gov.in/programmes/dilrmp
3	**Dharani — Telangana Integrated Land Records Management System**	dharani.telangana.gov.in
4	**CFMS — Comprehensive Financial Management System, Telangana**	cfms.telangana.gov.in
5	**Bhuvan — ISRO Geoportal & NIC GIS services**	bhuvan.nrsc.gov.in
6	**Survey of India / open district boundary datasets (GeoJSON)**	boundary geometry used in prototype — 33 districts of Telangana
7	**PM Gati Shakti National Master Plan**	gatishakti.gov.in
8	**e-Gazette — Gazette of India / Telangana State Gazette**	egazette.gov.in
9	**World Bank & NITI Aayog studies on acquisition-driven project delays**	problem-validation research (background study)
10	**SIH Problem Statement 26016 (official)**	MoRD · DoLR · Theme: Smart Automation · Category: Software

Prototype dataset note: boundaries from open Survey-of-India-derived GeoJSON; parcel/family records are realistic synthetic data — production links to Dharani / Bhuvan / CFMS.